

Course: ECE 5200

EXAM 2 Problems Retake

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Problems to be regraded: 2a 2b 3a 3b 3c

Revised Solutions

$$2.a) h[n] = \left(\frac{1}{2}\right)^n (u[n] - u[n-8])$$

$$h[n] = \begin{cases} (1/2)^n, & 0 \leq n \leq 7 \\ 0, & \text{otherwise} \end{cases}$$

$$\hat{h}[k] = 14 \left(e^{j2\pi k/N} \right), \quad k = 0, 1, \dots, N-1$$

$$\hat{h}[\bar{n}] = \sum_{m=-\infty}^{\infty} h[n-mN]$$

$$N=4$$

$$n: 0, 1, 2, 3$$

$$\hat{h}[\bar{n}] = h[\bar{n}] + h[\bar{n}+4]$$

$$\hat{h}[\bar{0}] = h[\bar{0}] + h[\bar{4}] = 1 + \frac{1}{16} = \frac{17}{16}$$

$$\hat{h}[\bar{1}] = h[\bar{1}] + h[\bar{5}] = \frac{1}{2} + \frac{1}{32} = \frac{17}{32}$$

$$\hat{h}[\bar{2}] = h[\bar{2}] + h[\bar{6}] = \frac{1}{4} + \frac{1}{64} = \frac{17}{64}$$

$$\hat{h}[\bar{3}] = h[\bar{3}] + h[\bar{7}] = \frac{1}{8} + \frac{1}{128} = \frac{17}{128}$$

$$\hat{h}[\bar{n}] = \begin{cases} \frac{17}{16}, & n=0 \pmod{4} \\ \frac{17}{32}, & n=1 \pmod{4} \\ \frac{17}{64}, & n=2 \pmod{4} \\ \frac{17}{128}, & n=3 \pmod{4} \end{cases}$$

$$\hat{h}[\bar{n}] \neq h[\bar{n}]$$

$$b) N=8$$

$$\hat{h}[\bar{n}] = \sum_{m=-\infty}^{\infty} h[n-mN]$$

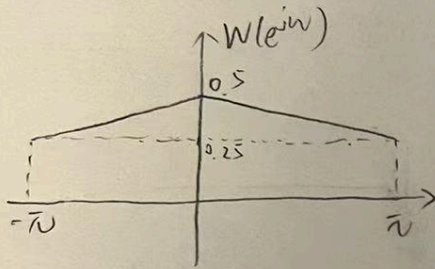
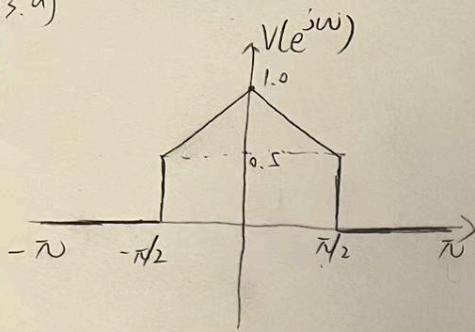
$$\hat{h}[\bar{n}] = h[\bar{n}], \quad 0 \leq n \leq 7$$

$$\hat{h}[\bar{n}] = \begin{cases} (1/2)^n, & 0 \leq n \leq 7 \\ \text{extension, otherwise} \end{cases}$$

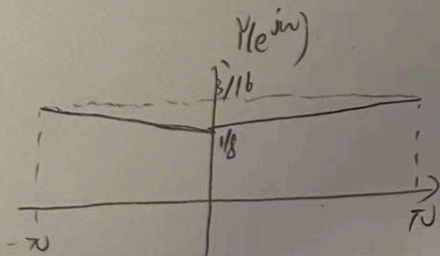
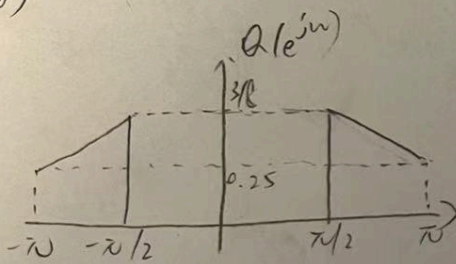
$$\text{in one period: } \hat{h}[\bar{0}] = 1, \quad \hat{h}[\bar{1}] = \frac{1}{2}, \quad \hat{h}[\bar{2}] = \frac{1}{4}, \quad \hat{h}[\bar{3}] = \frac{1}{8}$$

$$\hat{h}[\bar{4}] = \frac{1}{16}, \quad \hat{h}[\bar{5}] = \frac{1}{32}, \quad \hat{h}[\bar{6}] = \frac{1}{64}, \quad \hat{h}[\bar{7}] = \frac{1}{128}$$

3. a)



b)



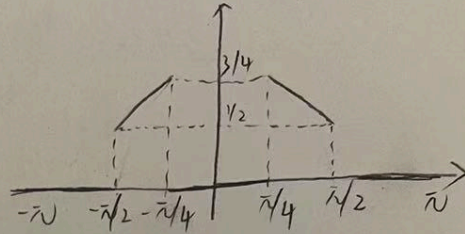
c) $H_1(z)$ with $\downarrow 2 \Rightarrow H_1(z^2)$

$x[n] \rightarrow G(z) \rightarrow \downarrow 4 \rightarrow y[n]$

$$G(z) = H_0(z) H_1(z^2)$$

$$H_1(z) = 1 - H_0(z)$$

$$G(z) = H_0(z) (1 - H_0(z^2))$$



Original Copy

Midterm #2

Friday, April 3, 2026

Name: Yangziquan Guo
ID: 500675783

Score: 45

Problem	Pts.	Score
1	20	14.5
2	20	5
3	30	22.5
Total	70	45

I certify that this is my sole work and I haven't discussed any problem with other person.

Signature: Yangziquan Guo

1. Please print your name and student ID number.
2. Sign on the above statement.
3. For full credit, you should show all of your work and explain your reasoning.

GOOD LUCK!

17.5

Problem 1 (20 Pts) (a) (10 Pts) Let $h[n]$ be an FIR filter given by

7.5
$$h[n] = -\frac{1}{2}\delta[n] + \delta[n-1] - \frac{1}{2}a_2\delta[n-2].$$

Let $g[n]$ be obtained through its z -transform satisfying $G(z) = -H(-z)$

where $H(z)$ denotes the z -transform of $h[n]$. Express $g[n]$ and determine whether it satisfies the generalized linear phase (GLP) property.

$$H(z) = -\frac{1}{2} + z^{-1} - \frac{1}{2}z^{-2}$$

$$G(z) = -H(-z) = -\left(-\frac{1}{2} + (-z)^{-1} - \frac{1}{2}(-z)^{-2}\right)$$

$$= \frac{1}{2} + z^{-1} - \frac{1}{2}z^{-2}$$
 mistake broke the symmetry

let $z^{-1} = s$

$$\left. \begin{aligned} \frac{1}{2}s^2 + s - \frac{1}{2} &= 0 \\ s^2 + 2s - 1 &= 0 \\ (s-1)^2 &= 0 \end{aligned} \right\} ??$$

centered at $s=1 \Rightarrow$ symmetry $g[n]$ satisfy GLP

(b) (10 Pts) Let $F_1(z) = (1+z^{-1})(1+\frac{1}{2}z^{-1})$ and $F_2(z) = (1-z^{-1})(1-\frac{1}{2}z^{-1})$. Determine whether the product

$$F_3(z) = F_1(z)F_2(z)$$

has the generalized linear phase (GLP) property.

$$F_3(z) = (1+z^{-1})(1+\frac{1}{2}z^{-1})(1-z^{-1})(1-\frac{1}{2}z^{-1})$$

zeros:
$$\begin{array}{cc} -z^{-1} & -\frac{1}{2} \\ -\frac{1}{2}z^{-1} & -\frac{1}{2z} \\ z^{-1} & \frac{1}{z} \\ \frac{1}{2z^{-1}} & \frac{1}{2z} \end{array}$$

these pairs neither conjugate or reciprocal

$F_3(z)$ Not GLP

$$-z^{-1} \rightarrow -z \text{ or } z$$

$$-\frac{1}{2}z^{-1} \rightarrow 2z \text{ or } -2z$$

Problem 2 (20 Pts) Let $h[n]$ be given by

$$h[n] = \left(\frac{1}{2}\right)^n (u[n] - u[n - M]), \quad M = 8,$$

where $u[n]$ denotes the unit-step function.

Let $\hat{h}[n]$ be obtained by first sampling the DTFT $H(e^{j\omega})$ on a length- N uniform frequency grid and then applying the inverse DFT:

$$\hat{H}[k] = H(e^{j\omega}) \Big|_{\omega = \frac{2\pi k}{N}}, \quad k = 0, 1, \dots, N-1.$$

(a) (10 Pts) Derive an explicit expression for $\hat{h}[n]$ when $N = 4$. Does $\hat{h}[n]$ coincide with $h[n]$? You may use the following facts:

$$\frac{1}{N} \sum_{k=0}^N \hat{H}[k] e^{j2\pi kn/N} = \frac{2\pi}{N} \cdot \frac{1}{2\pi} \int_{[0, 2\pi)} Z(e^{j\omega}) H(e^{j\omega}) e^{j\omega n} d\omega$$

where

$$Z(e^{j\omega}) = \sum_{k=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi k}{N}\right),$$

and the inverse DTFT of $Z(e^{j\omega})$ is

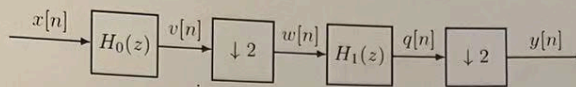
$$z[n] = \frac{N}{2\pi} \sum_{m=-\infty}^{\infty} \delta[n - mN].$$

$N=4, \omega: \frac{\pi k}{2}$
 $\hat{H}[k] = H(e^{j\pi k/2})$
 $\frac{1}{4} \sum_{k=0}^3 \hat{H}[k] e^{j2\pi kn/4} = \frac{1}{4} \int_0^{2\pi} H(e^{j\omega}) e^{j\omega n} \sum_{k=-\infty}^{\infty} \delta(\omega - \frac{2\pi k}{4}) d\omega$
 $h[n] = \delta[n] + \frac{1}{2} \delta[n-1] + \dots + (\frac{1}{2})^7 \delta[n-7] \leftarrow ??$
 $H(e^{j\omega}) = \sum_{d=0}^7 (\frac{1}{2})^d e^{-j\omega d}$
 incomplete

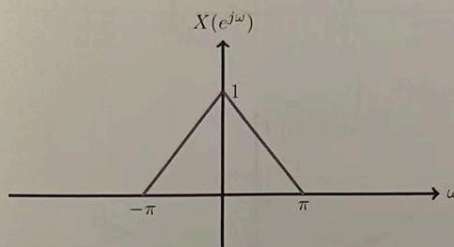
(b) (10 Pts) Repeat Part (a) with $N = 8$. Does $\hat{h}[n]$ coincide with $h[n]$ in this case?

22.5

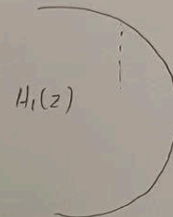
Problem 3 (30 Pts) We consider the following system.



Let $H_0(z)$ be an ideal lowpass filter of cutoff frequency $\pi/2$ and gain 1. Let $H_1(z) = 1 - H_0(z)$. Suppose that the DTFT of $x[n]$ is given as in the following plot.



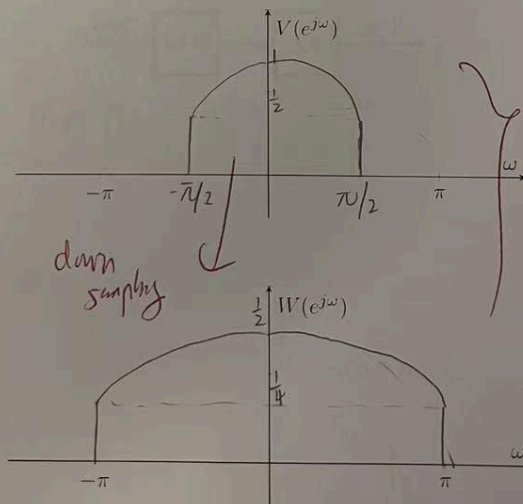
$H_0(z)$ low pass



(a) (10 Pts) Draw the DTFTs of $v[n]$ and $w[n]$. Label all graphs for full credits. You may use the fact that

$$W(e^{j\omega}) = \frac{1}{2} \left(V(e^{j\omega/2}) + V(e^{j(\omega-2\pi)/2}) \right).$$

1.5

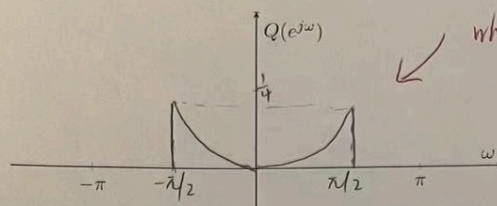


wrong graph

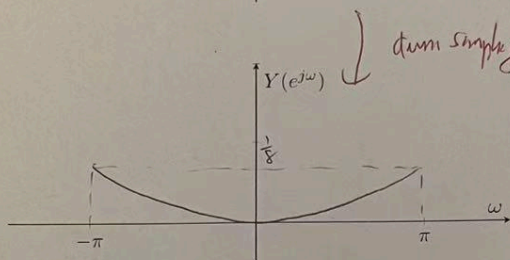
down sampling

- (b) (10 Pts) Draw the DTFTs of $q[n]$ and $y[n]$. Label all graphs for full credits.

7.5



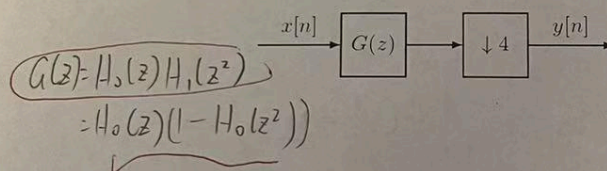
where did it
come from?
not constant
with (a)



down sample

- (c) (10 Pts) Find $G(z)$ given in terms of $H_0(z)$ and $H_1(z)$ such that the following block diagram provides an equivalent system:

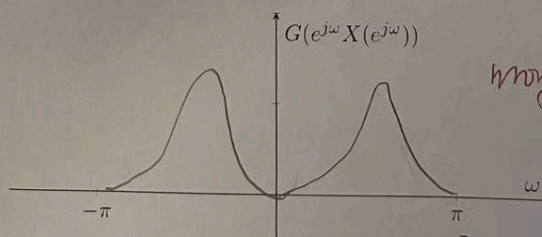
7.5



$$G(z) = H_0(z)H_1(z^2)$$

$$= H_0(z)(1 - H_0(z^2))$$

Draw $G(e^{j\omega})X(e^{j\omega})$ for $\omega \in [-\pi, \pi)$.



many
graph

label?